Quiz-2 topic

Q1

ALU is an integrated part of Bus Interface Unit (BIU). Do you agree? State your statement.

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No, I do not agree. In 8086, the ALU is part of the Execution Unit (EU), not the Bus Interface Unit (BIU). The BIU mainly handles address generation, instruction fetching, segment registers, IP and instruction queue. The EU contains the general registers, flag register, instruction decoder and ALU, which performs arithmetic and logical operations.

Q2

Whats the purpose of PC in an 8086 Microprocessor?

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In 8086, the program counter (PC) is called the Instruction Pointer. Its purpose is to hold the offset address of the next instruction to be fetched.

Q3

How does 8086 differentiate a memory and I/O operation?

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The 8086 differentiates memory and I/O operation by High and Low signals.

M/IO = 1 -> Memory operation

M/IO = 0 -> I/O Operation

Q4

Show with an example how a 20-bit physical address is made in 8086 stack operation.

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In stack operation, 8086 uses SS and SP.

Let,

SS = 0A00 H

SP = 0100 H

So,

$$\begin{array}{lr} \text{Physical address} = & 0A000 \\ & +\ 0100 \\ \hline & 0A100\ H \end{array}$$

Therefore, stack address,

SS:SP = 0A00H:0100H

Physical address = 0A100 H

Here,

SS = 16 bit

SP = 16 bit

And thus how a 20 bit physical address is made.

Q5

How does 8086 react if any unusual operation occurs like overflow, carry or division by zero.

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The 8086 reacts through the flag register and interrupt system.

After an arithmetic operation, the flag register is updated to show the processor status. For Carry, the 8086 normally does not stop the program automatically. It sets CF=1 and programmer can check this using conditional Jump instructions.

For overflow or division by zero, the interupt occurs and handled throught iterupts.

Q6

During an arithmetic operation, you need to generate an output at port X. How shall 8086 react and which pins shall be used?

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First, the 8086 performs the arithmetic operation inside ALU and EU. The result is usually kept In a register such as AL or AX and the flag register is updated. Then the result can be sent to port X using Pins like ALE, $M/\overline{IO}$, $\overline{WR}$, DEN, DT/R, READY, CLK.